

scRNA-seq Analysis Report

260429-pbmc3k

Project name: 260429-pbmc3k
Tissue / dataset type: N/A
Species: N/A
Date: 2026-04-29
Results directory: K:\@scRNA\results\260429-pbmc3k
Figures directory: K:\@scRNA\figures\260429-pbmc3k

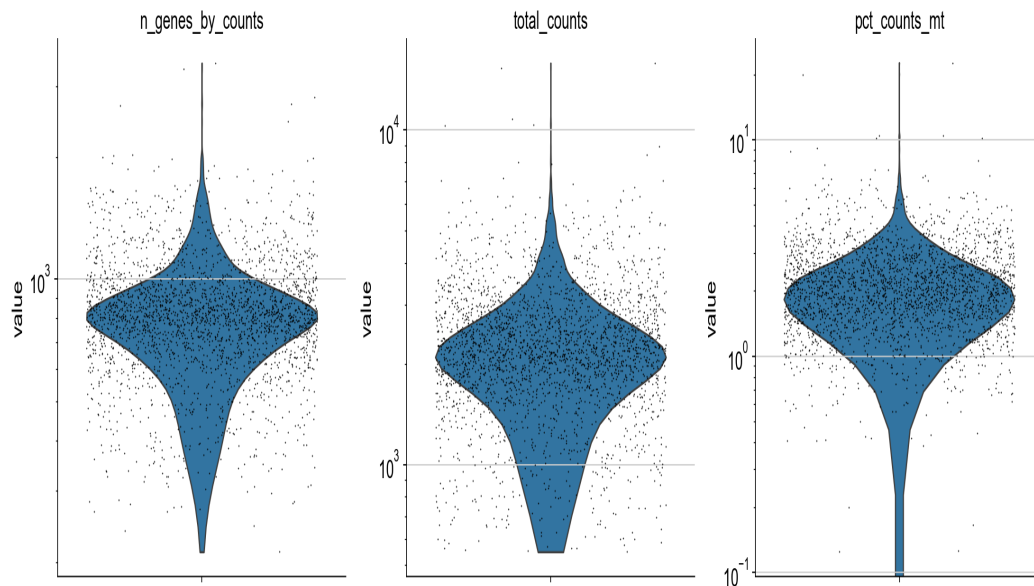
1. Pipeline Overview

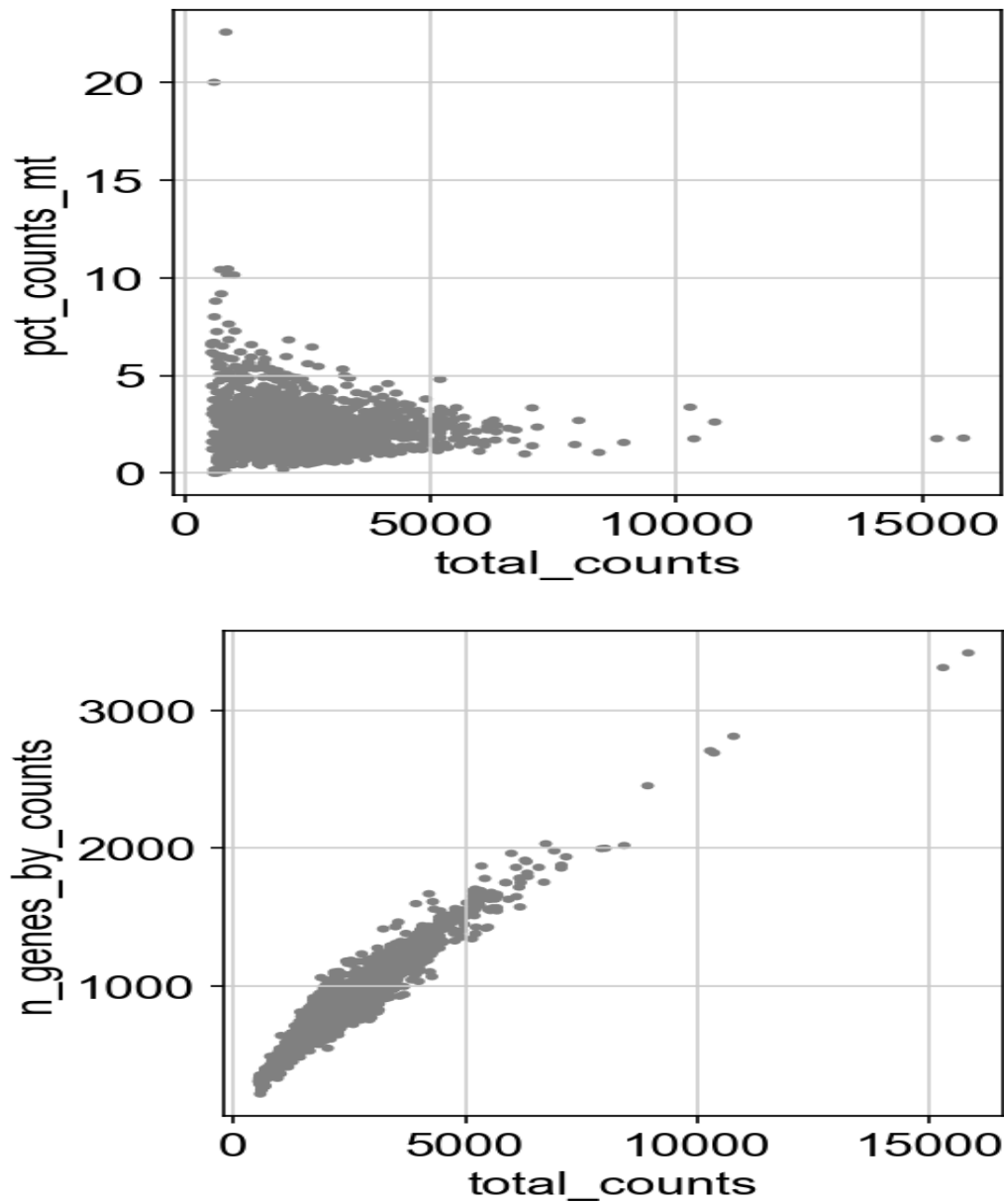
This report summarizes a standardized single-cell RNA-seq analysis pipeline, including: quality control, normalization and log-transformation, dimensionality reduction (PCA and UMAP), graph-based clustering (Leiden), marker-gene analysis, automatic cell-type annotation based on reference markers, and supplementary evaluation of annotation confidence and cluster consistency.

2. Quality Control (QC)

QC parameters (from YAML)

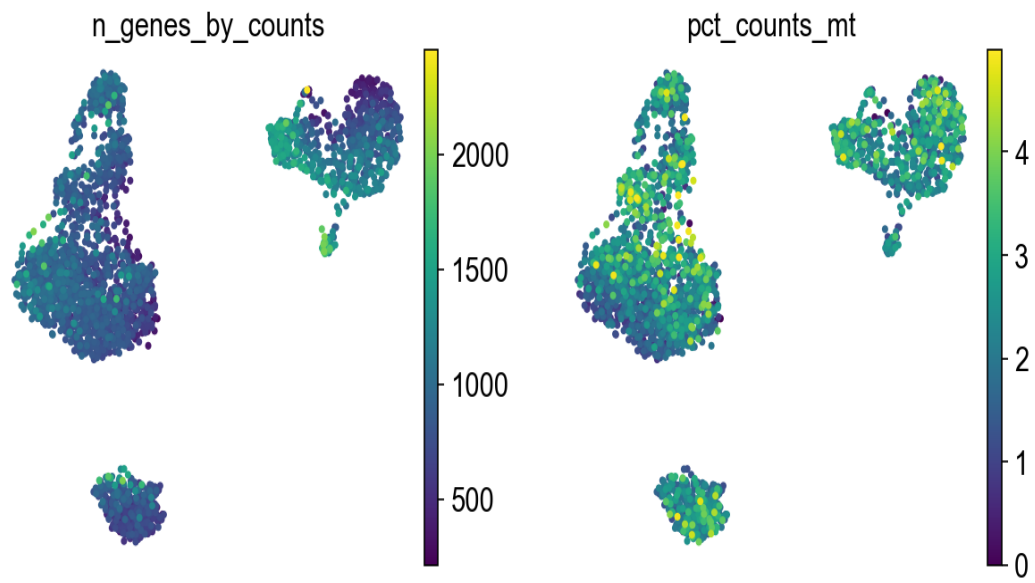
min_genes_per_cell: 200
max_genes_per_cell: 2500
max_pct_mt: 5.0
min_cells_per_gene: 3



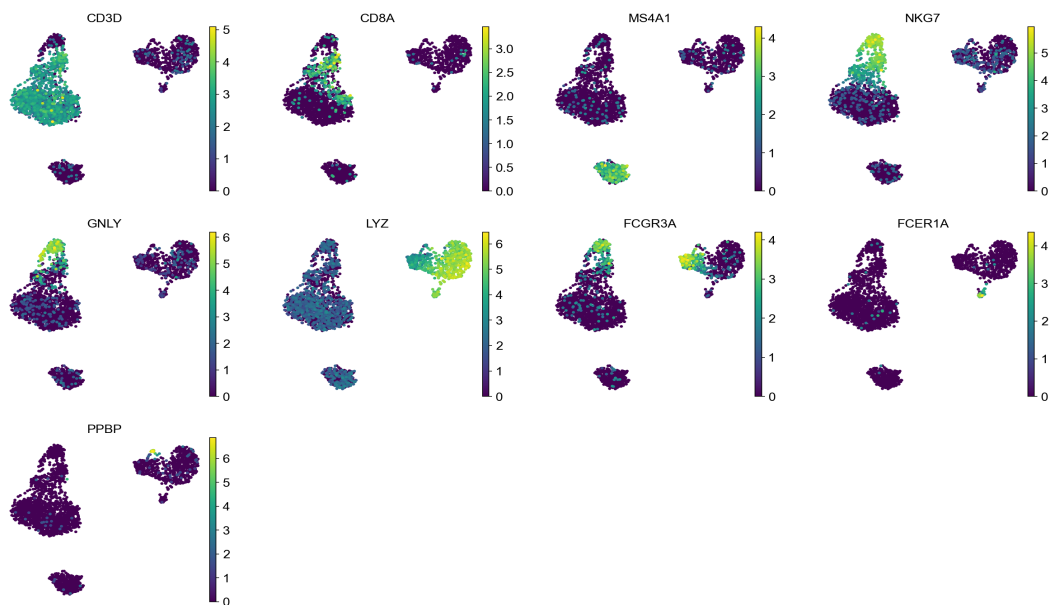


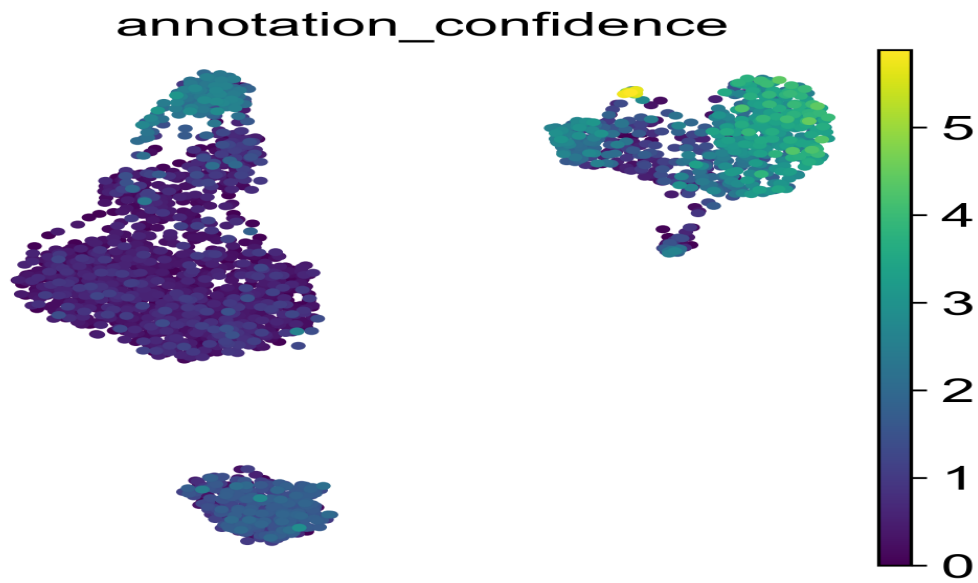
3. UMAP Embeddings & Clustering

UMAP embeddings provide a 2D representation of the high-dimensional transcriptomic space. Cells are colored by QC status, clusters, or cell types to illustrate global structure and major cell populations.



leiden





4. Cell-Type Annotation & Marker Genes

Automatic cell-type annotation was performed using 9 reference marker sets defined in the YAML configuration. Each cell was assigned the label whose marker gene score was highest.

cluster	gene	logFC	pval_adj
0	CCL5	5.80	1.4e-138
0	NKG7	5.24	2.6e-113
0	B2M	0.80	6.7e-81
1	CD74	4.09	5.2e-179
1	CD79A	7.61	2.0e-163
1	HLA-DRA	4.88	2.6e-162
2	LDHB	2.73	2.3e-226
2	RPS12	1.03	1.3e-214
2	RPS25	1.16	1.9e-194
3	LYZ	6.22	1.1e-233
3	S100A9	7.37	7.1e-233
3	S100A8	7.50	1.8e-225
4	NKG7	6.84	3.9e-90
4	GNLY	7.90	1.8e-85
4	GZMB	7.64	4.1e-82
5	COTL1	3.52	1.8e-92
5	LST1	4.87	4.4e-92
5	FCER1G	4.55	6.4e-90
6	HLA-DPB1	4.65	4.9e-16
6	HLA-DPA1	4.39	8.6e-16
6	HLA-DQA1	5.13	2.9e-15

5. Cluster vs Cell-Type Consistency

cluster	dominant cell type	n_cells	fraction_dominant
0	CD8_T	293	0.39
1	Naive_B	340	0.97
2	CD4_T	1159	0.55
3	CD14_Monocytes	451	0.97
4	NK_cells	161	0.98
5	FCGR3A_Monocytes	202	0.75
6	Dendritic_cells	32	0.84

6. Key Analysis Parameters (from YAML)

QC

min_genes_per_cell: 200
max_genes_per_cell: 2500
max_pct_mt: 5.0
min_cells_per_gene: 3

Analysis

n_top_hvgs: 2000
n_pcs: 40
neighbor_k: 15
leiden_resolution: 0.5

Annotation

reference marker groups: 9
Total cells: 2638
Total genes: 2000